

Maths Homework Template

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Short Math Guide: <http://tug.ctan.org/info/short-math-guide>.

Short Math Guide (Chinese version): <https://github.com/hoganbin/short-math-guide>.

1. ¶ Inline equations for simple math: $p = \hbar k$, $\varepsilon = \hbar \omega$, $\phi = \frac{1 + \sqrt{5}}{2}$.

¶ Centered equations for more involved or important equations:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i \quad (1)$$

$$\frac{d}{dx} e^{a(x)} = a'(x) e^{a(x)} \quad (2)$$

$$\frac{\|\mathbf{Ax}\|_\infty}{\|\mathbf{x}\|_\infty} = \frac{\max_i \left| \sum_{j=1}^n A_{ij} x_j \right|}{\max_i |x_{i1}|} \quad (3)$$

2. ¶ An example of a matrix:

$$\mathbf{M} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \\ a_1^2 & a_2^2 & \cdots & a_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \cdots & a_n^{n-1} \end{bmatrix}, \quad \det(\mathbf{M}) = \prod_{1 \leq j < i \leq n} (a_i - a_j) \quad (4)$$

¶ An example of accents:

$$\hat{h} \quad \check{c} \quad \tilde{t} \quad \acute{a} \quad \grave{g} \quad \dot{d} \quad \ddot{d} \quad \ddot{\ddot{d}} \quad \ddot{\ddot{\ddot{d}}} \quad \breve{b} \quad \bar{b} \quad \vec{v} \quad (5)$$

¶ An example of math fonts:

$$\begin{array}{cccc} ABCD & ABCD & \mathbf{ABCD} & ABCD \\ \mathcal{ABCD} & \mathcal{ABCD} & \mathfrak{ABCD} & ABCD \end{array} \quad (6)$$

¶ An example of roots:

$$r = \sqrt[3]{\frac{3}{1 + \sqrt{\frac{3}{1 + \sqrt{3}}}}} \quad (7)$$

¶ An example of Dirac notation:

$$\begin{array}{lll} \langle \phi | \psi \rangle & \left\langle \frac{1}{2} \lambda \middle| \psi \right\rangle & \left| \phi_{\hat{Q}=0} \right\rangle \\ \langle A \rangle & \left\langle \frac{1}{2} \lambda \right\rangle & \langle \Psi | A | \Psi \rangle \end{array} \tag{8}$$

3.

$$\iiint \nabla \cdot \boldsymbol{A} \, \mathrm{d}V = \oiint \boldsymbol{A} \, \mathrm{d}\boldsymbol{S}$$

4.

$$F_n(z)=\begin{cases}\frac{1}{\sqrt{1+in^2\mathrm{e}^{Az}}}, & z<0\\ \frac{1}{\sqrt{1+n^2\mathrm{e}^{Az}}}, & z\geqslant 0\end{cases}$$

5.

$$u(x,t)=\sum_{n=0}^{\infty}X_n(x)Y_n(y)=\frac{8a^2}{\pi^3}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^3}\left(1-\frac{\cosh\frac{2k+1}{a}\pi y}{\cosh\frac{2k+1}{2a}\pi b}\right)\sin\frac{2k+1}{a}\pi x$$

6.

$$\begin{aligned}\mathscr{L}\left\{\frac{1-\cos\omega t}{t}\right\}&=\int_p^\infty\left(\frac{1}{p}-\frac{1}{2(p+\mathrm{i}\omega)}-\frac{1}{2(p-\mathrm{i}\omega)}\right)\mathrm{d}t\\&=\ln p-\frac{1}{2}\ln(p+\mathrm{i}\omega)-\frac{1}{2}\ln(p-\mathrm{i}\omega)\bigg|_p^\infty\\&=\frac{1}{2}\ln\frac{p^2+\omega^2}{p^2}\end{aligned}$$

7.

$$\begin{aligned}[q,p]_{Q,P}&=\frac{\partial q}{\partial Q}\cdot\frac{\partial q}{\partial P}-\frac{\partial q}{\partial P}\cdot\frac{\partial q}{\partial Q}\\&=1+\frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}}-\frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}}\\&=1\end{aligned}$$

8.

$$\begin{aligned}\frac{4\pi R_\odot^2\cdot\sigma T_\odot^4}{4\pi d^2}&=1352.83\,\mathrm{W/m^2}\\T_\odot&=5763.43\,\mathrm{K}\\\lambda_\odot&=\frac{b}{T_\odot}=5027.86\,\mathrm{\AA}\end{aligned}$$

9.

$$\begin{aligned}I(x)&=\int_{-\frac{b}{2}}^{\frac{b}{2}}\mathrm{d}I(x)\\&=4\alpha\frac{I_0}{b}\int_{-\frac{b}{2}}^{\frac{b}{2}}\cos^2\left(\frac{\pi d}{\lambda D}\left(x+\frac{D}{R}\xi\right)\right)\mathrm{d}\xi\\&=4\alpha I_0\left(1+\frac{1}{u}\sin u\cos\left(\frac{2\pi d}{\lambda D}x\right)\right),\quad u=\frac{\pi db}{\lambda R}\end{aligned}$$